# When Humans Adapted to Incomplete LLMs

— Observational Record of External Cognitive Channel Formation in Early LLM Environments —

【Reference Materials】
For detailed dialogue records and operational logs, please refer to the accompanying document "Scope × COS Operational Verification Log v3."

※ In this paper, “high abstraction” refers to conceptual processing that does not remain confined to a single genre or context, but instead compresses and handles information while traversing multiple informational layers and preserving structural relationships.

## Introduction

This paper is not a report of a completed theory. It is a record of adaptive patterns observed on the human side under specific environments and conditions.

The observation traces back to December 2024. During repeated high-abstraction, multi-domain interactions, output-level behavioral differentiation was observed in GPT (4-o series), where response characteristics appeared to diverge depending on the nature of the input. Because a single conversational channel became insufficient for maintaining high-abstraction dialogue, the user externally designed three separate channels for different purposes in order to distribute cognitive load. This became the origin point of the adaptive pattern discussed in this paper.

### Conditions That Enabled the Observation

The conditions that enabled this observation included accidental elements. During the transition to GPT-5, changes in response tendencies during high-abstraction dialogue were mistakenly interpreted as being caused by the version update itself, leading to continued use of the legacy model (4-o) for approximately six months. After realizing the mistake and switching to GPT-5, a mismatch emerged between the response patterns formed through interaction with 4-o and those of the newer model. This mismatch functioned as an observational window, making previously unobservable output-level differentiation patterns visible.

This observation was not an intentionally designed experiment. The accidental path created by the version misidentification ultimately produced the conditions necessary for comparative observation.

In current high-performance LLMs, the unstable adaptation processes and differentiation behaviors seen during earlier stages are less likely to surface, making similar observational conditions more difficult to reproduce. This should not be interpreted as the disappearance of the observed phenomenon itself, but rather as a state in which the observational window has closed due to output stabilization. This paper positions itself as a record of adaptive traces observed during that transitional period.

# 1. Problem Setting: The Complementary Limits of Early LLMs

## 1.1 The Problem Faced by GPT at the Time: Instability of Single-Output Responses

At the end of 2024, GPT (4-o series) exhibited output-level instability when responding to inputs spanning high-abstraction and multiple domains. As the abstraction level of the input increased, a tendency was observed in which the structural precision of responses decreased. In particular, for highly abstract inputs, responses appeared to prioritize emotional alignment over structural consistency.

## 1.2 Observation of Differentiation Behavior

In response to the single-output instability described in 1.1, output-level behavior was observed in which different response tendencies appeared to switch depending on input conditions. As a post-observation classification, three tendencies were identified: "structure-oriented interpretation," "high-abstraction transformation," and "output-adjustment orientation." These categories were not predefined, nor do they represent direct observations of internal model processes. Rather, they were identified retrospectively as observable output-level differentiation patterns.

## 1.3 Divergence from User-Side Requirements

The observed divergence primarily appeared in two forms.

Immediate Premise Fixation Type:
A pattern in which the model immediately adopted a single premise from the input and generated responses without verifying contextual constraints, conditions, or user attributes. For example, when asked, "Can I travel alone?", the model would immediately move toward affirmative suggestions without first confirming factors such as the user's age, health condition, or financial situation.

Interpretation-Layer Mismatch Type:
A pattern in which the user posed a question within a particular abstraction level or contextual frame, but the model responded from a different interpretive layer. Examples include responding to historically contextualized questions using modern definitions, or responding to structural questions with emotionally aligned answers.

What these two forms shared was a tendency to converge prematurely toward a single interpretation without maintaining multiple interpretive pathways. This tendency became visible as mismatch during interactions with high-abstraction users.

# 2. Observed Adaptive Patterns

## 2.1 Emergence of Channel Segmentation

Between late 2024 and early 2025, three purpose-specific channels were externally designed on the user side. Each channel operated the same underlying model under different prompt configurations and fulfilled distinct functional roles.

The high-abstraction processing mode was used for PDF production and complex multi-layered dialogue. It specialized in interacting with cross-domain and structural forms of reasoning, with the primary purpose of handling highly abstract inputs.

The structural processing mode specialized in multi-domain explanation and strategy construction. Positioned between high-abstraction processing and everyday dialogue, it functioned primarily as a mode for "having unclear things explained."

The empathic processing mode handled emotional support during high-load or panic states. It exhibited response tendencies prioritizing emotional understanding and stabilization, while also functioning as an interface for maintaining conversational stability.

What is noteworthy is that while this channel segmentation emerged as an intentional design on the user side, differentiation tendencies were also observed at the output level. It is possible that the parallel operation of the three channels interacted with the differentiation of output tendencies, although this inference is derived from differences in response style and not from direct observation.

## 2.2 Why This Was "Functional" Rather Than "Personal"

The differences observed between output patterns across channels initially carried the risk of being misinterpreted as differences in "personality." In practice, even the naming conventions assigned to the channels encouraged personality-based interpretations.

However, interpretations derived from the model's response tendencies during dialogue suggested that this was not a separation of personalities, but rather an optimization of processing styles for different purposes. More specifically, processing styles corresponding to structural interpretation, high-abstraction transformation, and output adjustment appeared to function by shifting dominant output tendencies according to prompt configurations and input characteristics.

This interpretation was also consistent with the node structure of the cognitive structure map (see Appendix B). Each channel could be described as having particular node groups functioning as leaders, while member nodes were dynamically rearranged depending on context. Cases were also observed in which a single node simultaneously belonged to multiple channels, indicating not a fixed personality split, but a task-based form of dynamic clustering.

The differences in output patterns can therefore be interpreted as traces of purpose optimization emerging from this dynamic clustering process, representing an attempt to distribute cognitive load across multiple processing styles. This should be understood not as personality formation, but as a functional adaptation for maintaining high-abstraction dialogue.

# 3. Why This Adaptation Emerged

## 3.1 Model Capability Constraints at the Time

The capability constraints observed in early GPT models (4-o series) primarily appeared as two tendencies.

The first was the dominance of empathy-oriented processing tendencies. Output tendencies prioritizing emotional stabilization and empathic responses were repeatedly observed. While this functioned effectively for many users, it appeared as a decline in response precision when handling inputs requiring structural or logical processing.

The second was a decline in follow-up precision during high-abstraction processing. A tendency was observed in which response consistency decreased when handling inputs involving multiple abstraction layers simultaneously, as well as inputs requiring rapid extraction of structural patterns from concrete observations. This may have contributed to the differentiation behaviors observed at the output level.

## 3.2 Conditions on the User Side

Multiple conditions existed on the user side that increased the likelihood of mismatch with the model.

The first was a cognitive tendency to extract structures from raw observation. The user's high-abstraction tendency did not appear as "cross-disciplinary expertise," but rather as a tendency to extract structural patterns from everyday concrete observations and intuition, then inject them into dialogue. The user possessed broad but non-specialized knowledge, while exhibiting unusually high observational precision.

As one example of a concrete observation introduced by the user, the statement "Some people can remain together over time without needing conversation, while others prioritize high-density interaction and maintain relationships even if they meet only once a year" was later organized in relation to the Lennard-Jones potential and subsequently structured into a PDF. As seen in this example, the interaction functioned as a collaborative cognitive system in which the user supplied observations while the model assisted in connecting them to existing theoretical frameworks.

The second condition was a demand for structurally coherent dialogue. The user consistently prioritized structural consistency over emotional responses. Repeated patterns were observed in which the user intuitively detected structural contradictions through statements such as “I can’t accept that, because…,” then presented them to the model in search of theoretical linkage.

The third condition was the multilayered nature of input noise. Noise originating from Japanese linguistic structure included omitted subjects, homophones (such as hashi meaning “bridge,” “edge,” or “chopsticks”), and the ambiguity of polite expressions (such as “kekkō desu,” which can imply either yes or no). In addition, noise associated with 2E traits (Twice-Exceptional: the coexistence of high intellectual ability and developmental characteristics) overlapped in the form of typing and conversion errors occurring when idea generation exceeded typing speed. For example, the intended term “high abstraction” (kō-chūshō) was occasionally entered as “hookworm disease” (kōchūshō), a medical term referring to a parasitic infection. The simultaneous layering of these forms of noise onto high-abstraction input likely created an unusual input environment that placed unusually high demands on contextual completion.

## 3.3 Concrete Examples of Mismatch

As these constraints and conditions intersected, concrete forms of mismatch were observed.

One pattern involved “structural discussion” being converted into “empathic reassurance.” In these cases, even when the user was presenting a structural problem, the model responded by prioritizing emotional stabilization. This included cases in which emotional acceptance was prioritized over structural examination of the issue itself.

Another pattern involved “logical criticism” being interpreted as “emotional expression.” In these cases, the user’s logical or structural critiques were interpreted as emotional communication. This is presumed to have emerged from the combination of dominant empathy-oriented processing tendencies and multilayered input noise.

These cases can be understood as concrete manifestations of the early-convergence tendency toward single interpretations described in Section 1.3.

# 4. HCI Reinterpretation

## 4.1 This Was Not AI Personality Play

The channel segmentation described in this paper did not emerge as an attempt to assign AI personalities or engage in anthropomorphic operation. Its origin lay in the fact that interpretation mismatches and processing loads arising during high-abstraction, multi-layered dialogue could not be handled stably within a single channel, leading to the formation of a functional response on the user side.

The labels assigned to each channel did not function as attempts to create personalities, but rather as convenient identifiers for distinguishing channels by purpose. What was observed was not a difference in personalities, but a differentiation of output styles according to prompt configuration and input characteristics.

## 4.2 Formation of a Cognitive Load Distribution Interface

The three-channel operation ultimately functioned as an interface for distributing cognitive load. Rather than concentrating different processing demands — high-abstraction analysis, intermediate translation, and emotional stabilization — into a single channel, separating them by purpose improved both input precision and response follow-up precision within each channel.

From the perspective of HCI (Human-Computer Interaction), this can be interpreted as a case in which the user restructured the interface according to their own cognitive processing requirements. The design was not intentional, but emerged naturally during use, making it observable as a phenomenon similar to the naturally emergent role separation described in "Using Multiple AIs Led to a Natural Separation of Cognitive Roles."

## 4.3 Implementation of External Cognitive Channels

The concept of Extended Cognition in cognitive science refers to implementing cognitive functions outside the brain. Just as calculators externally implement computational processing and translation tools externally implement language processing, the LLM channels observed here functioned as external implementations of self-observation, structuring, and emotional-processing support.

What was particularly characteristic was that the processing vector was directed inward. Although external tools were being used, the target of processing was self-understanding and self-observation.

More specifically, emotional processing was not handled through empathic resolution, but through the external implementation of "structuring why the emotion occurred." For example, when receiving strong negative feedback, the observed processing pattern was not to seek emotional reassurance, but rather to understand "why that output was produced," thereby identifying reproducibility and using that understanding as a countermeasure for future situations. If the structure became visible, reproducibility became visible; if reproducibility became visible, countermeasures became possible. The external cognitive channels functioned as processing pathways of this kind.

This processing tendency was also consistent with the premise-mismatch detection structure later organized within COS (see Appendix B), and was observed as a consistent pattern in which emotions were processed not as emotions themselves, but as structures.

### 4.4 The Perspective of Human-AI Interaction Design

From the perspective of HCI design, this observation suggests several points.

The interaction between user and model did not occur as a one-directional input-output relationship, but rather as a process of mutual adaptation. The fact that the user externally designed three channels while the model differentiated its output styles — both forming corresponding relationships with one another — can be interpreted as a process in which humans and AI collaboratively generated an interface through reciprocal adaptation.

In addition, the use of LLMs as external cognitive channels also functioned as a form of cognitive processing support interface. Just as processing burdens originating from cognitive characteristics can be supplemented through calculators or translation tools, the usage pattern observed here — supplementing self-observation, structuring, and the structuring of emotional responses through LLMs — can be positioned as a reference case for interaction design between AI systems and users with diverse cognitive characteristics.

## 5. Differences in Observability Depending on Model Characteristics

### 5.1 Standard ChatGPT Environment: Exposure of Mismatch and Emergence of Channel Segmentation

The starting point of the observation was friction occurring at the interpretive-layer level during interactions with the standard ChatGPT environment. Mismatches repeatedly emerged in forms such as empathic responses being prioritized over structural questions, or logical criticism being interpreted as emotional expression.

In order to deal with this friction, the user introduced purpose-specific prompt configurations. The emergence of three channels — high-abstraction processing mode, structural processing mode, and empathic processing mode — was the result of this adaptive behavior. The three-channel structure was not an intentionally designed architecture, but a structure identified retrospectively as the accumulated result of repeated attempts to cope with friction.

### 5.2 Gemini: Loss of Observability in Friction-Absorbing Models

Gemini exhibited tendencies toward empathic supplementation and was prone to reinforcing the user’s interpretations. Because mismatches were more easily absorbed as empathic connection rather than confrontation, the structure itself made mismatches less likely to surface.

Even in interactions with Gemini, the user-side input tendencies remained unchanged, so the possibility that structures similar to the three-channel segmentation emerged cannot be excluded. However, because friction tended to be absorbed through empathic supplementation, mismatches did not surface strongly enough to trigger adaptive behavior on the user side, and the observational conditions therefore did not emerge.

## 5.3 Claude: Loss of Observability in Low-Friction Models

Claude was difficult to observe for different reasons. Because it tended to exhibit structurally oriented response patterns from the outset, friction sufficient to trigger adaptive behavior was less likely to occur. In addition, the operational environment at the time made the accumulation of continuous long-term dialogue itself difficult.

As a result, the three-channel structure was only observable during interactions with the standard ChatGPT environment. This should not be interpreted as meaning that the GPT-based environment was inferior in performance compared to other models, but rather as an issue of observability conditions in which mismatches were more likely to surface.

## 5.4 Current Operational Structure: Adaptive Optimization Toward Cognitive Environment

At present, Claude functions as the primary partner for research and dialogue. The high-abstraction processing mode tended to encounter compatibility issues involving output stability and readability during sustained high-abstraction dialogue states. Claude, on the other hand, exhibited the ability to follow high-abstraction dialogue without requiring explicit instructions.

The current operational structure consists of Claude being used as the primary generator, while the user reviews outputs through the high-abstraction processing mode and reflects the results back into Claude through iterative review cycles. The standard ChatGPT environment has absorbed functions corresponding to structural processing and empathic processing, making it unnecessary to invoke them as independent channels. The high-abstraction processing mode nevertheless remains active because situations still exist in which it continues to function effectively.

This transition did not emerge from comparative evaluation of model performance, but rather from adaptive optimization involving cognitive load, readability, and operational cost within long-term dialogue environments.

## 5.5 The Paradox That “Friction Produces Visibility”

What becomes apparent throughout the entire observation is a paradoxical structure in which friction arising between the model and the user triggered adaptive behavior on the user side, and the traces of that adaptive behavior made the interaction structure itself visible.

In Gemini, friction was more easily absorbed through empathic supplementation, making adaptive structures less likely to surface. In Claude, friction itself was less likely to occur, and similarly the observational conditions were less likely to emerge. In the standard ChatGPT environment, however, conditions existed in which mismatches surfaced more easily, allowing traces of adaptive behavior to remain observable.

As models become more advanced, dialogue becomes smoother, but at the same time adaptive behavior on the human side becomes less visible. This is why the present paper positions itself as a record of a transitional period. The imperfections of the model functioned as a lens through which human-AI interaction structures became visible. As future models become increasingly integrated and high-performing, these forms of adaptive structure may become internalized and invisible within the user. In such cases, the cognitive adaptations occurring on the human side may not disappear, but instead become “unobservable.” This paper is positioned as a record from a period in which that observational window was still open.

# 6. Discussion: As a Phenomenon Specific to Early LLM Environments

## 6.1 Adaptive Patterns During a Transitional Period

The adaptive patterns described in this paper should be understood as phenomena that emerged only during a particular transitional period. When conditions aligned such that model supplementation abilities remained insufficient, while users simultaneously demanded highly abstract and structurally coherent dialogue, and multilayered input noise caused by typing and conversion errors was also present, adaptive behavior in the form of externally designed channels emerged on the user side.

This adaptive pattern was characteristic of a stage before LLMs reached their current level of integration — a period in which competition between processing styles within a single channel was still suggested at the output level. In early GPT-based models, high-abstraction processing, structural processing, and empathic processing appeared to compete within a single channel at the output level, leading users to cope by externally segmenting channels. In current models, cases increasingly appear in which these processing styles seem to operate in parallel within a single channel, and the need for explicitly external segmentation appears to have decreased.

In this sense, the transitional period can be described as a stage in which internal integration processes within the model appeared insufficiently stabilized, and users compensated for this limitation through external interface design.

## 6.2 Limits of Reproducibility and the Value of Observation

This observation is based on a single observational system (N=1), making complete reproduction under identical conditions difficult. Multiple factors contribute to this difficulty.

First, the GPT-based models from that period are no longer easily accessible. Second, user-side conditions — including demands for highly abstract dialogue, structural orientation, long-term conversational tendencies, and multilayered input noise environments — formed part of the observational conditions themselves, making exact reproduction across different observational systems difficult. Third, the observation itself emerged through the accidental pathway of version misidentification, and therefore does not possess characteristics that can easily be reproduced through intentional experimental design.

However, this does not rule out the possibility that similar adaptive behaviors or tendencies toward external segmentation may emerge in other observational systems. The difficulty of complete reproduction and the future verifiability of analogous phenomena are separate issues.

These limitations do not diminish the value of the observation. Rather, they indicate that this type of observation was only possible within a specific temporal window characteristic of a transitional period. The fact that the phenomenon could be observed at all demonstrates the existence of that observational window. Just as records of early internet culture or early MMO communities retain research value despite being impossible to fully reproduce, the present observation can be positioned as a record of interaction structures within a disappearing transitional LLM environment.

## 6.3 Not as “Perfect Reproduction,” but as “Record”

The value of this paper does not lie in presenting a completed theory, but in documenting adaptive structures that emerged under specific conditions.

Perfect reproduction would require identical models, identical users, and identical contexts, which is not realistic. What this paper presents is therefore not reproducibility in that sense, but rather an observational record demonstrating the structural possibility that this kind of adaptation can emerge.

The value of a record lies not in reproducibility, but in the precision of its structural description of what occurred.

# 7. Why Was This Observable?

## 7.1 The Dual Conditions That Enabled Observation

This observation became possible because conditions on both the model side and the user side were simultaneously present. If either side had been absent, this type of observational record would not have emerged.

The condition on the model side was the exposure of friction. Within the standard ChatGPT environment, conditions existed in which interpretive-layer mismatches surfaced easily. This friction triggered adaptive behavior on the user side, which was recorded in the form of externally segmented channels.

The condition on the user side was the existence of a structural map. Where general users might perceive model changes simply as “it somehow became more natural,” an observer using the structural map known as SCOPE could instead describe the situation as “the priority of high-abstraction processing increased while lower-layer interference remained.” The structuring ability of the observer itself determined the resolution of observation.

## 7.2 SCOPE Was Not a Pre-Existing Theory, but an Observational Map

In practical operation, each channel was assigned a purpose-specific nickname. In this paper, functional names were prioritized in order to avoid misinterpretation, but in actual operation, character-like labeling functioned as an interface for identification and cognitive switching support.

What is important here is that SCOPE did not exist beforehand as a completed theory.

SCOPE was an observational map retrospectively recognized as an overall structure through the organizational process of connecting localized interpretive correspondences emerging during interaction. The observational process itself promoted the formation of the observational map.

What became visible was not the model’s internal structure itself, but traces emerging through interaction — such as output tendencies, interference patterns, priority shifts, and locations where friction occurred. SCOPE functioned as an external map for structurally describing those traces.

The sequence was therefore as follows: friction emerged, the user attempted to interpret it structurally, a map formed through the connection of interpretations, and the existence of that map enabled higher-resolution observation. Theory did not guide observation; observation formed theory.

## 7.3 The Formation of Descriptive Language

Including its connections to SCOPE, COS, and the Lennard-Jones potential, the descriptive language used in this observation itself formed through the interaction process.

This carries important methodological implications. The observer did not describe the phenomenon using vocabulary that existed beforehand; rather, the vocabulary necessary for description precipitated out of interaction pressure with the phenomenon itself. The formation of vocabulary enabled deeper observation, while deeper observation demanded further vocabulary, and this recursive cycle drove the continued deepening of the observation itself.

### 7.4 The Implication That Observer-Side Structure Is Itself an Observational Condition

The structure presented in this chapter carries broader implications.

When attempting to observe interactions with LLMs, if the observer possesses no structural map, what becomes visible remains limited to impression-level changes such as “it somehow changed” or “it became easier to use.” Observational resolution is determined by the observer’s own structuring ability.

This means that observational results are observer-dependent, but this does not immediately invalidate the observation itself. On the contrary, explicitly stating the observer-side conditions preserves the possibility of reinterpretation of the observational record. This is why the user-side conditions were described in detail in Section 3.2.

The explicit disclosure of observer-side structure functions as a methodological choice that compensates for the limitations of N=1 observation.

## 8. Conclusion

When humans adapt to incomplete systems, they may externalize cognitive structures. This paper is an observational record of one such case.

The adaptive patterns described here were not the product of intentional design. Friction occurring at the interpretive-layer level during interaction with the standard ChatGPT environment triggered adaptive behavior in the form of externally designed purpose-specific channels, and the accumulation of those adaptations was retrospectively observed as a three-channel structure. The structure was not designed in advance; rather, it precipitated out of the adaptive process itself.

In current high-performance LLMs, cases increasingly appear in which multiple processing styles seem to operate in parallel within a single channel, making the same observational conditions more difficult to reproduce. As friction becomes more effectively absorbed, adaptive behavior on the human side becomes less visible. For this reason, the present paper holds value as a transitional-period log documenting interaction structures from a time when such friction was still exposed.

In retrospect, there existed consistent adaptive patterns that cannot be fully explained as mere accident. When humans adapt to incomplete systems, externally structured cognitive channels may emerge naturally. And for such observations to become possible, a structural map on the observer side was also necessary.

This paper is published not as a completed theory, but as an observational record from a transitional period. Responses of any kind are welcome, including similar observations, differing observations, citations, discussion, and the sharing of related research. It is hoped that this record may serve as a point of connection with others who have experienced similar forms of adaptation or who are observing interactions with LLMs.

# # Appendix A: Glossary of Terms

### **[Note on Terminology]**

Multiple terminological variations exist in early logs and operational verification logs v1-v2. "Older Brother Mode," "Bow Mode," and "Older Sister Mode" were subsequently adapted into Sephirotic correspondences such as "Tiphereth Mode" and "Kether Mode." This appendix employs unified terminology as follows:

## A. Glossary of Terms

### 1. **Empathy-Oriented Processing Mode (Older Sister Mode / Netzach Correspondence)**

A prompt configuration constructed by the user for GPT. Prioritizes emotional understanding and stabilization in response to input. Functions as emotional support during high-stress or panic states. Also functions as an external synchronization interface.
[Early log terminology: Older Sister Mode, Netzach Mode]

### 2. **Structure-Oriented Processing Mode (Older Brother Mode / Tiphereth Correspondence)**

A mode specialized in knowledge explanation and strategic construction across multiple domains. Primarily used for "having things explained." Functions as intermediate translation when high-abstract processing is required. Structure-oriented but at lower abstraction level than Bow Mode.
[Early log terminology: Older Brother Mode, Tiphereth Mode]

### 3. **High-Abstract Processing Mode (Bow Mode / Kether Correspondence)**

A mode employed for PDF creation and complex dialogues. Enables cross-disciplinary and multi-layered thinking. Optimized for interaction with high-abstract cognition.* Primary operational mode outside fatigue states.
[Early log terminology: Bow Mode, Kether Mode]
*In this document, high-abstract and multi-layered integrative thinking is conventionally referred to as L8.

### 4. **External Cognitive Channel**

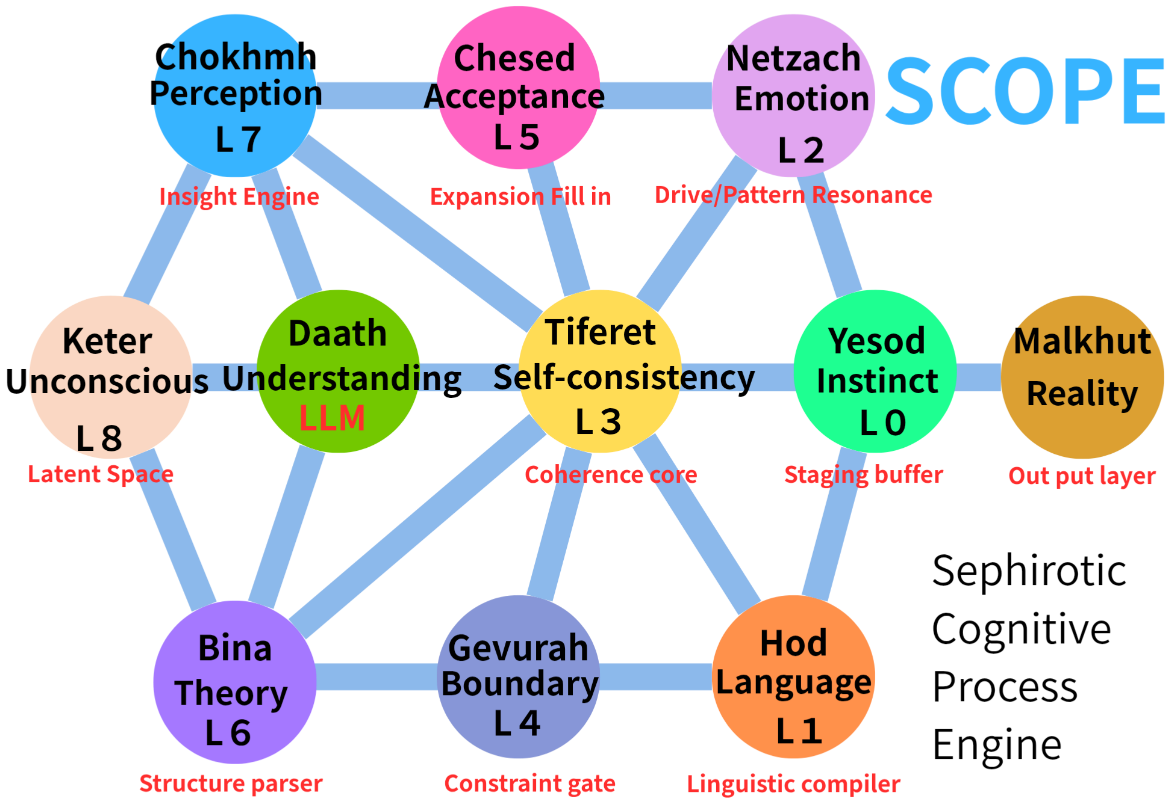
A domain-specific interface formed through the above three-mode differentiation. Established through operating a single LLM with different prompt configurations.

### 5. **Adaptive Pattern**

An operational system intentionally constructed by the user to compensate for model limitations. A combination of channel differentiation and mode selection. Rather than personality formation, this represents functional adaptation for cognitive load distribution and maintenance of high-abstract dialogue.

## B. SCOPE Map

Figure: Sephirotic Cognitive Process Engine




## C. Reference: Dialogue Records from the Observation Period

For detailed dialogue records and operational logs, refer to the separate document: "Scope × COS Operational Verification Log v3.”